NEXUS · DOCUMENTATION PACKAGE · DELIVERABLE 03

Application Flow

Navigation map, key user journeys, system interactions and error/alternate flows.

Document control

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Contents

1. 1 Navigation map
2. 2 Key user journeys
3. 3 System interactions
    1. 3.1 Offline save and upload (SEQ-01)
    2. 3.2 Read with cache fallback (SEQ-02)
    3. 3.3 Authentication (SEQ-03)
4. 4 Error and alternate flows

1 Navigation map

Routes are type-safe `@Serializable` objects in `NexusNavHost.kt`. The app starts at `Splash`. An auth guard sends any signed-out user on a protected route to `Login`, and sends a signed-in user on Login/Register to `Dashboard` Implemented. A bottom navigation bar (Today, Projects, Habits, Settings) appears on those four top-level routes only. A sync status banner appears on every authenticated screen except Splash.

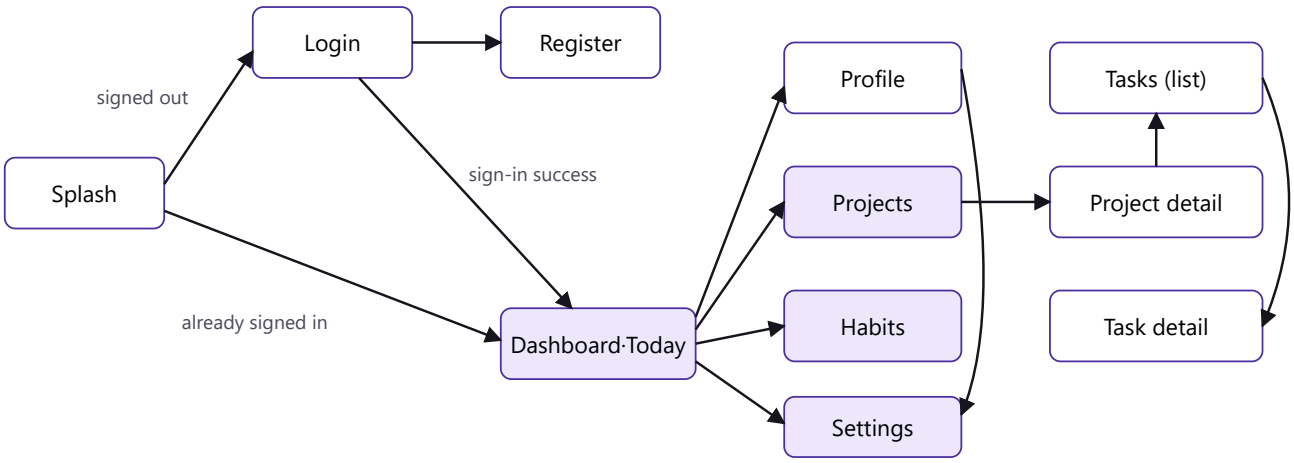


Figure 2. Navigation map. Shaded boxes are bottom-bar destinations. Back navigation pops the stack.

2 Key user journeys

Each journey lists the requirements it exercises. Steps reflect the implemented UI labels.

User journeys

ID	Journey	Main path	Requirements
UF-01	First launch and registration	Splash animation, then Login, then "Need an account? Register". Enter Display Name, Email and Password, tap Create Account. A JWT is saved and the Dashboard opens with an empty state, "Start your first project".	FR-01, FR-19
UF-02	Returning user	Splash, then Dashboard directly (token present). The Dashboard refresh uploads queued changes, then downloads projects and all their tasks into Room.	FR-02, FR-09, FR-16
UF-03	Plan a project	Projects tab, then the + (New Project) button. Enter a name, create, and open the project. Use "Add Task" or "Open Tasks Screen", then + (New Task): set title, due date, priority, status, labels and checklist, and save.	FR-10, FR-11
UF-04	Decide what to do today	The Today tab shows the Overdue / Due today / Upcoming cards. In a project's task list, use search, the status chips or "High priority"; open a task for its detail view; mark it complete.	FR-09, FR-12
UF-05	Daily habit check-in	Habits tab, then + (Add habit), name it and choose a frequency. Tap "Complete today" each day; stats (level, points, streaks, badges) recalculate on the device.	FR-13, FR-14

ID	Journey	Main path	Requirements
UF-06	Work offline	With the API unreachable, create or edit items as usual. The banner reads "Saved on device · waiting to sync" with "*n* pending changes". When the network returns (network callback, the 30 s loop, or tapping Sync), the queue uploads in order and the banner shows "All changes synced".	FR-16, TR-04
UF-07	Resolve a conflict	An upload is rejected with 409 (or 400). The banner shows "An item changed on another device · review required" with a Review button. The dialog offers "Keep changes" (the queue stays blocked and the edits are kept) or "Discard item edits" (that entity's queued operations and its local row are removed, then the rest of the queue retries).	FR-17, AUD-19
UF-08	Session expiry during sync	Upload returns 401, the banner says "Sign in again to sync saved changes", and a "Sign in" button clears the token and opens Login. Queued changes stay on the device for the same account.	FR-02, FR-16
UF-09	Manage account	Settings tab: edit Display Name; choose Appearance (System/Light/Dark) and Language; toggle reminders; "Sync account"; "Export habits CSV", then "Share CSV"; "Change password"; "Delete account and data" (confirm); "Log Out".	FR-04 to FR-08, FR-15
UF-11	Sign in with Google (v2.0)	Login or Register: tap "Continue with Google", pick an account in the Credential Manager sheet. The app posts the ID token to `/auth/google`, receives a Nexus JWT and opens Today. First use creates the account; an existing email account is linked.	FR-21
UF-12	Notifications (v2.0)	Settings: keep "Task due-date reminders" on (grant permission on Android 13+). A reminder lists overdue and due-today tasks. "Send test notification" shows a local notification and reports the server push result. Editing on phone A pushes "Task … updated on another device" to phone B, which refreshes its cache.	FR-22, FR-23
UF-10	Shared device, account switch	User A signs out with pending changes and user B signs in. B sees only B's rows. A's queue is neither shown nor uploaded until A signs in again.	FR-03, NFR-03

3 System interactions

3.1 Offline save and upload (SEQ-01)

1. The ViewModel calls a repository write, for example `TaskRepository.createTask()`. The repository generates a UUID for new entities on the device.
2. `SyncManager.save()` takes the mutex, then in one Room transaction writes the entity row and a `pending_changes` row (`operationId`, `resource`, `entityId`, `action`, JSON `payload`).
3. Later, `flushLocked()` reads the account's pending rows by `sequence` and posts each one to `POST /api/sync`. A row is deleted only after a 2xx response.
4. Server: if the `operationId` is already in the user's receipts it returns 204. Otherwise it validates the payload against the resource schema, checks for conflicts (update of a missing item, duplicate create, or a task whose project does not exist for this user), applies the change, records the receipt, saves atomically and returns 204. If the save fails, the in-memory change is rolled back.
5. The first failure stops the loop. Its HTTP status decides the banner message and whether Review is offered (400/409).

3.2 Read with cache fallback (SEQ-02)

1. `SyncManager.read()` takes the mutex and flushes pending changes first.
2. If the flush fails, Room data is returned, so unsent edits are never overwritten by server data.
3. If the flush succeeds, the remote call runs and replaces the account's cached rows in one transaction. On `IOException` the cache is returned and the banner reads "Offline · showing saved data". Other HTTP errors (for example 401) are passed up to the ViewModel.

3.3 Authentication (SEQ-03)

1. Login/Register posts credentials; the API checks them with bcrypt and returns a JWT with `sub` = user id, valid for 24 h.
2. `AuthSession.saveToken()` stores it encrypted. The cache owner id is decoded from the `sub` claim (not verified on the device; used only to scope the cache).
3. The OkHttp interceptor adds `Authorization: Bearer` to every Nexus API request.

4 Error and alternate flows

How failures surface to the user

ID	Condition	System behaviour	User sees
EF-01	API unreachable or timed out (5 s connect, 15 s call)	Writes are queued; reads fall back to Room.	Banner "Offline · changes saved on device" or "Offline · showing saved data"; Dashboard "Showing saved data". Sign-in shows "No internet connection. Check your network and try again."
EF-02	401/403 on a screen request	`ApiError.SessionExpired`; Settings clears the session.	"Your session has expired. Please log in again." Since v2.0, a wrong password at sign-in shows "Email or password is incorrect." instead (AUD-18 fixed).
EF-03	409 at registration (email exists) or 429 (rate limit)	Mapped by `AuthViewModel.authErrorMessage` (v2.0).	"An account with this email already exists. Sign in instead." / "Too many attempts. Please wait 15 minutes and try again."
EF-09	Google sign-in cancelled, no Google account on the device, or server SSO not configured	Cancel shows nothing; the other cases show a message.	"Add a Google account to this device…" / "Google sign-in is not available on this server yet."
EF-04	409/400 during sync	The queue stops at the failing operation; `conflict=true`.	Review button (UF-07)
EF-05	5xx from the API	`ApiError.ServerError`. In production the API hides internal messages.	"Something went wrong on our end. Please try again shortly." Sync banner: "Sync needs attention · saved changes kept on device".
EF-06	Item not available offline	Room has no row (never downloaded).	Error state with Retry, e.g. "Project is not available offline" (CON-05)
EF-07	Maintenance mode or announcement in `/app/config`	Dashboard banner. The API currently hard-codes maintenance off.	Maintenance message or the first announcement
EF-08	Delete account fails (offline or error)	The session is kept and nothing is purged.	Error message in Settings

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